

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (Biotechnology) (CBCS) Degree Examinations
 December 2023

BT2103-1 – MOLECULAR BIOLOGY & GENETIC ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- With respect to the human genome, which of the following is correct
 A) Contains 3.5 billion nucleotides
 B) Contains 3.5 billion pairs of nucleotides
 C) Contains 3.5 million pairs of nucleotides
 D) Contains 3.5 million nucleotides
- During DNA replication the function of Helicases is to:
 A) Separation of 2 strands of DNA
 B) Binding to the single strand of DNA
 C) Prevents torsion by DNA breaks
 D) Stabilises polymerase
- Which of the following proteins 'seal the nick via phosphodiester linkage' on a DNA molecule during replication?
 A) Catalases
 B) Ligases
 C) Topoisomerases
 D) Helicases
- DNA-dependent RNA synthesis is known as _____.
 A) Translation
 B) Reverse Transcription
 C) Transcription
 D) Teminism
- Which among the following organisms was used by Griffith to prove the concept of transformation?
 A) Staphylococcus pneumoniae
 B) Klebsiella pneumoniae
 C) Micrococcus pneumoniae
 D) Streptococcus pneumoniae
- Which among the following radio-isotopes was used in the Hershey - Chase experiment to radiolabel DNA?
 A) ^{32}P
 B) ^{32}S
 C) ^{35}P
 D) ^{14}N
- Primase is responsible for the _____ during DNA replication
 A) Separation of DNA strands
 B) the synthesis of DNA primer
 C) Synthesis of RNA primer
 D) Binding to single strand of DNA
- Which of the following leads to frameshift mutation?
 A) Alkylation
 B) Nucleotide Insertion
 C) Methylation
 D) Deamination
- Which among the following processes removes UV-induced thymine dimer formation?
 A) Direct Repair
 B) Base excision repair
 C) Nucleotide excision repair
 D) Recombination repair
- DNA gets transcribed by a protein known as:
 A) Primase
 B) DNA-dependent RNA polymerase
 C) DNA polymerase
 D) Helicase
- Protein coding regions of mRNA composed of a non-overlapping string of codons are called:
 A) Intron
 B) Codon
 C) Exons
 D) Open Reading Frame (ORF)
- The t-RNA can contain which one of the unusual bases?
 A) Hypoxanthine
 B) Methylated Adenine
 C) Methylated Guanine
 D) Hydroxylated Adenine
- mRNA will bind to which of the following?
 A) Anticodon loop of tRNA
 B) D loop
 C) Variable loop
 D) Acceptor arm

P.T.O.

14. Which of the following combinations is correct with respect to prokaryotic ribosomes?
 A) 50S+30S B) 50S+40S
 C) 40S+60S D) 30S+60S
15. The proper order of tRNA binding sites on the ribosomes is:
 A) P site, A site and E site B) A Site, P site and E site
 C) E site, P site and A site D) A site, E site, and P site
16. _____ class of restriction enzymes are used commonly in rDNA technology.
 A) Type I B) Type III
 C) Type II D) Type IV
17. Reverse transcriptase is found in which of the following organisms?
 A) HIV B) T2 Phage
 C) *E. coli* D) Mycoplasma
18. The function of Polynucleotide kinase is to _____.
 A) Join DNA fragments B) Remove phosphate group
 C) Add phosphatase group D) Cut the DNA
19. The self-DNA of bacteria is protected against their own restriction enzymes by the process of:
 A) Acetylation B) Methylation
 C) Deacetylation D) Carboxylation
20. DNA strand uncoiling is facilitated by _____ enzyme.
 A) Gyrase B) Polynucleotide kinase
 C) Phosphonucleotide kinase D) DNase

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) DNA is the transforming element – Justify using a detailed classical example.	10	L*2	1	2,12
	b) With an appropriate diagram, discuss the molecular organization of chromosome.	6	L2	1	2,12
2.	a) Discuss in detail various types of DNA damage. Add a note on its consequences.	10	L2	2	2,12
	b) Write notes on various types of models proposed to describe DNA replication.	6	L1	2	2,12
3.	a) Write short notes on: i) Karyotyping and its significance, ii) Complexity of eukaryotic genome.	8	L2	1	2,12
	b) Write notes on Nucleotide excision repair.	8	L2	2	2,12
Unit – II					
4.	a) Give a detailed account of Genetic Code.	8	L4	3	2,6
	b) Discuss in detail about post-transcriptional modifications.	8	L2	3	2,6
5.	a) With a suitable example, discuss the regulation of gene expression in prokaryotes.	10	L4	4	2,6
	b) Write a brief note on the activation of amino acids.	6	L2	4	2,6
6.	a) Write short notes on: i) Eukaryotic RNA Polymerase, ii) Transcription factors.	8	L2	3	2,6
	b) Give an account on the molecular events of protein synthesis.	8	L3	4	2,6
Unit – III					
7.	a) Define restriction enzymes. Elaborate the classification of restriction enzymes with suitable example.	8	L3	5	2,12
	b) With a suitable diagram, describe the steps involved in the generation of a recombinant DNA molecule.	8	L2	5	2,12
8.	a) Elaborate the physical modes of gene transfer.	10	L3	5	2,12
	b) Write short note on plasmids as vectors.	6	L2	5	2,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
